

Understanding Q-Learning and Deep Q-Learning in 2026:

A Methodological and Empirical Survey

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Abstract—Q-learning remains a cornerstone of Reinforcement Learning (RL), underpinning many state-of-the-art deep RL algorithms. Yet, despite its foundational status, no survey has systematically organized the diverse innovations that have shaped Q-learning over the past three decades. This paper fills that gap by presenting a comprehensive review of Q-learning variants, organized around the eight foundational weaknesses of vanilla Q-learning that modern methods address — overestimation bias, sample inefficiency, brittle exploration, reward sparsity and credit assignment, distribution shift, multi-agent coordination, slow adaptation, and function-approximation instability. We introduce a unified taxonomy covering both tabular and deep Q-learning methods, including offline RL, multi-agent value decomposition, and distributed Q-learning, and compile benchmark results from original papers across Atari and classic control environments. To complement the literature review, we provide a comparative analysis of six major open-source deep RL repositories — Tianshou, XuanCe, CleanRL, DQN Zoo, Stable Baselines3, and RLlib — highlighting their coverage of Q-learning algorithms and practical implementation challenges. Together, these contributions offer a structured resource for researchers and practitioners to navigate the evolution, empirical performance, and implementation landscape of Q-learning.

Index Terms—Deep Learning, Machine Learning, Reinforcement Learning, Q-Learning, Survey

IMPACT STATEMENT

Reinforcement learning increasingly underpins decision-making systems in robotics, autonomous vehicles, recommendation, and the post-training of large language models — and value-based Q-learning sits beneath much of this technology. Yet practitioners face a fragmented literature: dozens of algorithmic variants whose purpose, trade-offs, and software availability are scattered across three decades of papers. This survey reorganizes that landscape around the specific weaknesses each method was designed to fix, letting an engineer move directly from a symptom — overestimation, sparse rewards, instability at scale — to the methods that address it and the open-source libraries that implement them. By pairing this diagnostic taxonomy with calibrated benchmark evidence and a reproducibility audit of six widely used codebases, the work lowers the barrier to selecting, deploying, and extending Q-learning methods, and surfaces where research and tooling investment are most needed.

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I. INTRODUCTION

Reinforcement Learning (RL) and Deep Reinforcement Learning (DRL) have emerged as powerful paradigms for solving complex sequential decision-making problems, with applications ranging from Atari games [1], [2] and robotic control [3] to autonomous driving [4]. Among DRL approaches, two major families dominate: value-based methods, which estimate action-value functions, and policy-based methods, which directly optimize policy distributions.

While policy-based algorithms have shown strong performance in tasks with continuous action spaces, they often suffer from high variance and sensitivity to hyperparameters. In contrast, value-based methods such as Q-learning [5] and its deep variant DQN [1] remain widely used due to their conceptual simplicity, sample efficiency in discrete domains, and ease of implementation.

Q-learning’s longevity has produced a sprawling methodological literature. Over three decades, dozens of variants have been proposed to address specific limitations — overestimation bias, brittle exploration, sample inefficiency, instability under function approximation — and many have been combined into composite agents such as Rainbow [6]. Despite this volume of work, the *organizing structure* through which the field is presented has remained largely chronological or method-typed: distributional methods, ensemble methods, replay innovations, and so on. Existing surveys [7–10], as well as more recent broad-RL treatments [11], have inherited this method-type organization.

Method-type organization is bibliometrically convenient but analytically thin. It tells the reader *what kind of thing* a method is (an ensemble, a distributional model, a replay buffer modification) without surfacing *why* the method exists — which weakness of vanilla Q-learning it was designed to address, what trade-offs it introduces, and how it compares to other approaches that target the same weakness. As a result, readers encounter algorithms as isolated artifacts rather than as positioned moves in an ongoing technical conversation. Cross-method synthesis suffers accordingly.

This paper offers an alternative organization. Rather than presenting Q-learning’s variants by method type, we organize the field around **the foundational weaknesses of vanilla Q-learning that modern methods address**. Eight such weaknesses are identified:

1. *Overestimation bias* introduced by the max operator over noisy value estimates;

2. *Sample inefficiency* arising from uniform experience replay;
3. *Brittle exploration* under ε -greedy action selection;
4. *Reward sparsity and credit assignment* in long-horizon tasks;
5. *Distribution shift* when transferring to fixed-data (offline) regimes;
6. *Multi-agent coordination* failures when factoring single-agent Q across cooperative agents;
7. *Slow adaptation* when training per-task without transfer; and
8. *Function-approximation instability* — the “deadly triad” of off-policy learning, bootstrapping, and function approximation.

Each Related Works section in this paper corresponds to one of these weaknesses. Within each section, methods are grouped by the *mechanism* they use to address the weakness, compared head-to-head on the trade-offs they introduce, and evaluated against the empirical evidence relevant to that axis. The same method can appear in multiple sections when it advances multiple axes — Rainbow [6], for instance, recurs across five of the eight — and these cross-references form an explicit map of the field’s connective tissue.

This organization preserves all of the paper’s empirical and bibliographic contributions while reframing the surrounding analysis. Five distinguishing contributions, summarized in Table I, support the reframing:

1. A unified problem-axis taxonomy that integrates tabular Q-learning, classical deep Q-learning, and modern Q-based methods (offline RL, multi-agent value decomposition, distributed scaling) within a single framework.
2. A comparative analysis of six widely used open-source DRL repositories — Tianshou [12], XuanCe [13], CleanRL [14], DQN Zoo [15], Stable Baselines3 [16], and RLlib [17] — annotated *by axis* to reveal where the open-source ecosystem provides coverage and where it does not.
3. Curated benchmark results extracted from the Atari [2] suite as reported in the original papers, stratified by task category, and re-interpreted as evidence for or against each algorithmic axis rather than as a leaderboard.
4. Tabular Q-learning benchmark results from our own controlled reimplementations, isolating algorithmic design from architectural confound.
5. A thorough per-paper literature review organized by axis, with each method positioned by the weakness it addresses, the mechanism it uses, and the trade-offs it introduces.

The remainder of the paper is organized as follows. Section II introduces the MDP formalism and formally states the eight weaknesses that structure the Related Works. Section III describes the methodology. Section IV — the bulk of the paper — presents the problem-first review across the eight axes (§IV.A–H), followed by a ninth subsection §IV.I covering recent theoretical advances. Sections V and VI report Atari and tabular benchmark analyses. Section VII presents the repository comparison. Section VIII concludes with future direc-

tions, including a planned community-maintained Q-learning repository whose roadmap is informed by the gaps identified in Section VII. Supplementary material provides the full per-method mapping between the problem-first axis taxonomy of §IV and the method-type taxonomy of prior surveys, together with notation conventions and selected derivations.

Reading guide. The §IV body is written for mechanism and trade-off: each axis-section names the weakness, surveys the solution families that respond, and compares them on the costs they introduce. Equations in §IV are mechanism-defining rather than fully derived; longer derivations and proof sketches are concentrated in the supplementary material and may be skipped on a first read without losing the structural argument. §IV.I revisits the same ground theoretically.

II. BACKGROUND AND PROBLEM SETUP

II.A. Markov Decision Processes: We consider the standard formulation of a Markov Decision Process (MDP), defined as $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, R, P, \gamma, d_0 \rangle$, where $\mathcal{S}$ denotes the set of all possible states, and $\mathcal{A}$ the set of possible actions. The reward function $R : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ assigns a real-valued reward to each state-action pair. The transition function $P(s' | s, a)$ specifies the probability of transitioning to state s' from state s after taking action a . The discount factor $\gamma \in [0, 1)$ determines the importance of future rewards, and d_0 denotes the initial state distribution.

A policy $\pi : \mathcal{S} \rightarrow \mathcal{A}$ (or, in the stochastic case, $\pi : \mathcal{S} \rightarrow \Delta(\mathcal{A})$) prescribes an action for each state. The value of a policy at state s is the expected discounted return under that policy: $V^\pi(s) = \mathbb{E}_\pi [\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) | s_0 = s]$. The associated action-value function is $Q^\pi(s, a) = \mathbb{E}_\pi [\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) | s_0 = s, a_0 = a]$. Q-learning [5] estimates the optimal action-value $Q^*(s, a) = \max_\pi Q^\pi(s, a)$ via the recursive Bellman optimality equation,

$$Q^*(s, a) = \mathbb{E}_{s' \sim P(\cdot | s, a)} \left[r + \gamma \max_{a'} Q^*(s', a') \right],$$

implemented as an iterative update,

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left(r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right).$$

The remainder of this paper surveys methods that modify, augment, or replace components of this update to address one or more of the weaknesses introduced below.

II.B. Eight Weaknesses of Vanilla Q-Learning: The Q-learning update of §II.A is provably convergent under tabular representation and standard stochastic approximation conditions [Watkins & Dayan 1992]. In practical settings — function approximation, partial coverage, sparse rewards, multi-agent environments, fixed-data regimes — the update exhibits eight distinct failure modes. These failure modes are not independent; several share underlying mechanisms. But each motivates a recognizably different family of methods, and each is the organizing principle of one Related Works subsection.

W1. Overestimation bias. The operator $\max_{a'} Q(s', a')$ is biased upward in expectation whenever $Q(s', \cdot)$ contains zero-mean noise. Formally, for noisy estimators $\hat{Q}(s', a) = Q^*(s', a) + \epsilon_a$ with $\mathbb{E}[\epsilon_a] = 0$,

$$\mathbb{E} \left[\max_{a'} \tilde{Q}(s', a') \right] \geq \max_{a'} Q^*(s', a'),$$

with strict inequality whenever the noise is not degenerate [Thrun & Schwartz 1993]. Under recursive Bellman backups this bias propagates and amplifies, steering the greedy policy toward actions whose values are least accurately estimated. Methods responding to this weakness are surveyed in §IV.A.

W2. Sample inefficiency. Vanilla Q-learning uses each transition once and uniformly. The introduction of experience replay [18] allowed transitions to be reused, but uniform sampling treats all transitions as equally informative — a transition where the agent already predicts the outcome accurately contributes little to the update, yet is sampled as often as a high-error transition. In environments where high-information transitions are rare (sparse reward, low-probability state encounters), uniform replay yields slow convergence. Methods responding to this weakness — prioritized sampling, learning from demonstrations, memory-efficient replay — are surveyed in §IV.B.

W3. Brittle exploration. ε -greedy action selection takes a random action with probability ε and a greedy action otherwise. This suffices for environments where reward is sufficiently dense that near-greedy policies explore the state space through their own exploitation. In environments where rewards are sparse or delayed beyond an ε -greedy random-walk’s reach — Montezuma’s Revenge [2], Pitfall! [2], Private Eye [2] — ε -greedy exploration is dithered rather than directed and fails to escape early-state plateaus. The formal characterization is that ε -greedy is myopic with respect to posterior uncertainty in $Q(s, a)$; methods responding to this weakness inject structured noise, maintain posterior estimates, or use intrinsic motivation, and are surveyed in §IV.C.

W4. Reward sparsity and credit assignment. When reward is received only at the end of a long trajectory, the one-step Bellman backup requires many iterations to propagate the signal to early states. Multi-step returns [19] partially mitigate this by allowing n -step bootstrapping:

$$y_t^{(n)} = r_t + \gamma r_{t+1} + \dots + \gamma^{n-1} r_{t+n-1} + \gamma^n \max_{a'} Q(s_{t+n}, a').$$

But n -step returns trade variance for bias and do not address the deeper question of *what information* the return signal carries. Distributional RL [20] reframes the question: rather than estimating the expected return $\mathbb{E}[Z(s, a)]$, estimate the full distribution $Z(s, a)$. The distribution carries richer credit-assignment information — bimodality, skewness, tail behavior — that the scalar expectation discards. Methods responding to this weakness are surveyed in §IV.D.

W5. Distribution shift. Q-learning is off-policy *in principle*: the learned Q^* is independent of the behavior policy that generated transitions. In practice, the off-policy guarantee is fragile when the behavior policy is fixed and the support of the replay distribution is narrow — the regime of offline RL. The maximization $\max_{a'} Q(s', a')$ now ranges over actions a' that may be unsupported by the data, producing arbitrary extrapolation errors. The failure is structural: any algorithm that backs up through unsupported actions inherits unbounded

error. Methods responding to this weakness constrain Q -estimates or actions to remain within the data support and are surveyed in §IV.E.

W6. Multi-agent coordination. When the environment contains multiple cooperating agents with a shared reward, naive per-agent Q-learning treats other agents as part of the environment — a non-stationarity that breaks the MDP assumption. Centralized Q-learning over the joint action space avoids this but scales exponentially: $|\mathcal{A}|^N$ for N agents. Value decomposition methods factor $Q_{\text{tot}}(\mathbf{s}, \mathbf{a}) = f(Q_1(s_1, a_1), \dots, Q_N(s_N, a_N))$ under structural constraints on f (additivity, monotonicity) that preserve the *individual-global-max* property — that the action maximizing Q_{tot} jointly is the per-agent argmax of each Q_i . Methods responding to this weakness are surveyed in §IV.F.

W7. Slow adaptation and sample throughput (composite axis). Vanilla Q-learning is bottlenecked along two distinct but interrelated dimensions that we treat as a single axis. The *adaptation* dimension: vanilla Q-learning trains a single Q -function for a single task; transfer to a related task — different reward, different transition dynamics, different state distribution — requires retraining from scratch or initialization from a pre-trained Q . Neither is satisfactory when adaptation must occur online and within a small number of episodes. The *throughput* dimension: a single sequential learner is bottlenecked on the rate at which the environment produces transitions, even with experience replay. Both dimensions admit the same set of methodological responses — distributed actor-learner architectures, recurrent value functions, meta-learning approaches, and the recent predictable-scaling literature — because at scale the distinction between “scale to adapt across tasks” and “scale to learn faster on one task” blurs (Agent57 is the paradigm case). We treat them as one composite axis rather than two separate ones because the methodological responses are mostly shared; readers who prefer a finer-grained taxonomy can read §IV.G as covering two sub-axes W7a (sample throughput / distributed systems) and W7b (slow adaptation / meta-learning), which we surface explicitly in §IV.G.A. Methods are surveyed in §IV.G.

W8. Function-approximation instability. The combination of off-policy learning, bootstrapping, and function approximation — the *deadly triad* [21] — is provably unstable in the general case. Concretely, the iterates of $Q(s, a; \theta) \leftarrow Q(s, a; \theta) + \alpha(r + \gamma \max_{a'} Q(s', a'; \theta) - Q(s, a; \theta))$ need not converge, may diverge, and even when they converge can do so to a point that is far from Q^* . Target networks, Polyak averaging, layer normalization, and architectural decomposition each address different aspects of this instability. Methods responding to this weakness are surveyed in §IV.H.

II.C. Notation conventions: Throughout the paper, s, a, r, s' denote a transition; θ and θ^- denote the parameters of an online and target network, respectively; π denotes a policy; α a learning rate; γ a discount factor; $\mathcal{D}$ a replay buffer; and ϵ either an exploration rate or a noise variable, disambiguated by context. Deviations from this convention are explicitly flagged in each section.

III. METHODOLOGY

This review surveys Q-learning algorithms spanning over three decades of development, from foundational tabular approaches to recent deep variants. We include influential papers that shaped the theoretical landscape, introduced practical improvements, or demonstrated wide applicability in real-world domains. The selection prioritizes methods whose contributions can be mapped onto one or more of the eight weakness axes introduced in §II.B — methods that respond to a recognizable weakness of vanilla Q-learning with a distinct mechanism.

A. Source selection and review protocol: This paper is best characterized as a *narrative review with empirical add-ons* rather than a strictly systematic review in the PRISMA sense: the synthesis is interpretive (each axis carries an argued reading of the methods it surveys), and the eight-axis framework was developed iteratively with the literature rather than applied as a pre-registered classification. We document the search and selection protocol here so that the resulting axis assignments, benchmark extractions, and repository comparisons can be independently audited.

Databases and search strategy. Source candidates were identified from three databases: Google Scholar (broad coverage, citation-graph traversal), arXiv (cs.LG and cs.AI listings, primary source for preprints), and Semantic Scholar (citation-relationship analysis). The search ran from December 2025 through May 2026 with a cutoff date of 2026-05-15 for the late-2025 / 2026 paper sweep documented separately. Searches combined topical anchors (“Q-learning”, “deep Q-network”, “DQN”) with axis-specific terms (“overestimation bias”, “prioritized replay”, “exploration bonus”, “distributional reinforcement learning”, “offline reinforcement learning”, “value decomposition”, “distributed reinforcement learning”, “meta-reinforcement learning”, “function approximation stability”) and recency filters (publication date 2018-onward for deep-RL methods; full historical range for foundational results).

Inclusion and exclusion criteria. Five inclusion criteria were applied:

1. **Method centrality** — the work introduces a methodological contribution to the Q-learning family rather than applying Q-learning to a domain.
2. **Mechanism distinctness** — the contribution is mechanistically distinguishable from prior work along at least one of the eight axes of §II.B.
3. **Empirical validation** — the work reports results on a recognized benchmark (Atari, classic control, D4RL, SMAC, ProcGen, or equivalent).
4. **Cross-reference impact** — the work is cited as foundational by subsequent methods in the same axis-family, *or* the work was published within the previous twelve months and represents an active research thread.
5. **Reproducibility** — code is publicly available or the algorithm is sufficiently specified to be independently re-implemented.

Exclusion criteria removed: (a) application-domain papers using Q-learning as a black-box tool without methodological contribution; (b) workshop-only or non-archival preprints lack-

ing subsequent follow-up; (c) methods that have been demonstrably superseded by equivalent-effort alternatives (e.g., we cover Maximin Q-Learning but not all of its precursors). Methods satisfying all five inclusion criteria received per-paper treatment. Methods satisfying a subset received compact treatment as part of a family (e.g., the value-decomposition family in §IV.F is treated through its canonical members VDN, QMIX, QPLEX, QTRAN, with QFIX added on the basis of criterion 4’s twelve-month clause).

Screening flow. Initial candidate identification produced approximately 200 papers; first-pass title/abstract screening narrowed this to approximately 120. A subsequent full-text screening against the five inclusion criteria produced approximately 80 papers receiving full per-method treatment, with an additional 20 cited at family or cross-reference level. We do not publish a strict PRISMA flow diagram here because the search-and-iterate process was non-linear: as the eight-axis framework crystallized, several methods that initially appeared incidental were promoted to per-paper status, and conversely some once-canonical entries were demoted to family-level mention. We treat this iteration as a methodological *limitation* rather than a defect (and surface it explicitly in §VIII.D); a strictly systematic review would have required pre-registering the axis framework before the literature search, which we did not.

Axis-assignment protocol. Each method covered in §IV is assigned to a *primary axis* — the weakness whose response motivated the method’s introduction — and zero or more *secondary axes* — additional weaknesses the method incidentally addresses or partially mitigates. Primary assignment is made on the basis of the method’s stated motivation in its original publication and the predominant mechanism of its contribution. Where the original publication’s motivation differs from our axis assignment (notably for distributional methods in §IV.D and Dueling DQN in §IV.H), the re-interpretation is argued explicitly within the section. The complete per-method mapping, including secondary axes and cross-references, is provided as supplementary material.

Per-paper extraction template. For methods receiving per-paper treatment, we extracted: (i) the formal weakness statement the method addresses (§IV.X.A reference); (ii) the key mechanism- defining equation; (iii) the primary empirical evidence reported in the original publication; (iv) trade-offs explicitly discussed by the authors; (v) any subsequent ablations or follow-up evaluations relevant to the trade-off discussion. This extraction template is not published as supplementary material in the current version, but the per-method writeups in §IV reflect its structure consistently.

B. Comparison with prior surveys: To position this survey against existing work, we compiled Table I, which compares this paper with five prior Q-learning-focused surveys [7–10] and [11] along five distinguishing dimensions: analysis of public DQN repositories, unified taxonomy spanning tabular and deep Q-learning, extraction of original-paper Atari benchmarks, classic control benchmarks from controlled re-implementation, and per-paper review organized around mechanism and axis. Where prior surveys focus on chronological or method-type organization, this paper’s problem-first axis

structure (§IV) is, to our knowledge, the first multi-axis problem-first treatment of Q-learning specifically. This claim is supported by a documented 2024–2026 sweep across Google Scholar, arXiv (cs.LG and cs.AI), and Semantic Scholar that surveyed every recent Q-learning, DQN, and broad-RL survey we could identify; the closest prior art is the single-axis problem-first organization of [22], which addresses distribution shift in offline RL only. Search strings, dates, and per-survey notes are recorded as supplementary material so the claim can be independently audited.

C. Axis assignment methodology: Each method covered in this paper is assigned to a *primary axis* — the weakness whose response motivated the method’s introduction — and zero or more *secondary axes* — additional weaknesses the method incidentally addresses or partially mitigates. Primary assignment is made on the basis of the method’s stated motivation in its original publication and the predominant mechanism of its contribution. Where the original publication’s motivation differs from our axis assignment (notably for distributional methods in §IV.D and Dueling DQN in §IV.H), the reinterpretation is argued explicitly within the section.

Cross-references between axis-sections — e.g., Rainbow appearing in §IV.B (primary) and being cross-referenced from §IV.A, §IV.C, §IV.D, and §IV.H — surface the connective tissue that catalogue-style surveys obscure. The method-type taxonomy table in §IV gives the category-level mapping; a full per-method index supporting readers who arrive expecting the older organization is provided as supplementary material.

D. Empirical evidence sources: Section V analyzes Atari benchmark results extracted from the original papers introducing each method. Section VI reports tabular benchmarks from controlled re-implementations on Gymnasium environments (FrozenLake-v1, Taxi-v3, CliffWalking-v1). Section VII analyzes algorithmic coverage across six widely-used open-source deep RL repositories (Tianshou [12], XuanCe [13], CleanRL [14], DQN Zoo [15], Stable Baselines3 [16], RLlib [17]). The three evidence streams together support different aspects of the paper’s contribution: the literature analysis demonstrates methodological diversity, the controlled experiments isolate algorithmic from architectural effects, and the repository analysis maps the practical landscape of available implementations.

IV. RELATED WORKS

Over the past three decades, Q-learning and its deep variants have evolved through a long sequence of mechanism-level innovations. Section II.B identified eight foundational weaknesses of vanilla Q-learning that motivate the field’s trajectory; this section surveys the methods that respond to each. The organization is problem-first: each subsection corresponds to one weakness, groups responses by mechanism, and compares them on the trade-offs they introduce.

The method-type taxonomy. Prior Q-learning surveys [7–10] organize methods by *type* — what kind of mechanism a method is. We call this the *method-type taxonomy* and use that term throughout. The table below lists its six categories and, crucially, shows where each category’s methods fall among the eight problem axes. The categories cut *across* the

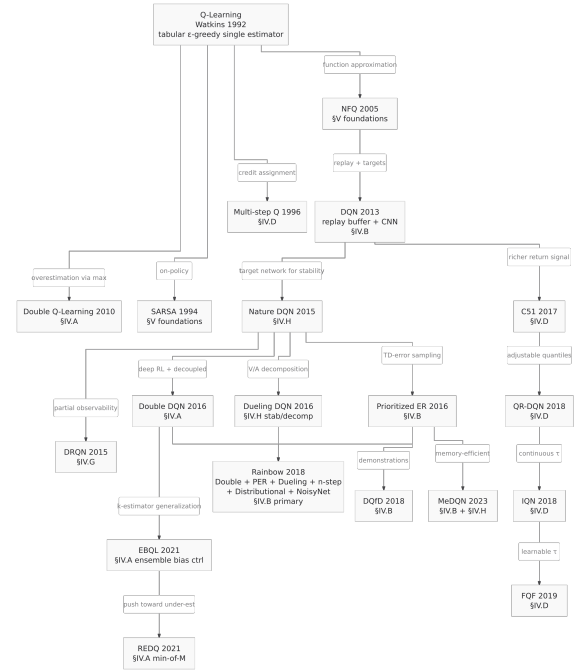


Fig. 1. Method genealogy of Q-learning and its deep variants. Edges are labeled by the weakness of the parent that the child method addresses.

axes: statistical methods split between exploration (§IV.C) and credit assignment (§IV.D), and ensemble methods between overestimation control (§IV.A) and exploration (§IV.C). A problem-first organization regroups exactly these cross-cutting cases by the weakness each method addresses rather than by what the method is. (The full per-method mapping is provided as supplementary material.)

This subsection orients the reader through three further artifacts: a method genealogy showing the inheritance relationships between Q-learning’s variants, a companion figure showing the modern-RL branches that have emerged outside the conventional taxonomy, and an axis × mechanism-family matrix summarizing what mechanism types are deployed against each weakness. The §IV subsections that follow each provide a five-part structure: formal statement of the weakness (subsection A), grouped solution families (subsection B), trade-offs (subsection C), empirical evidence (subsection D), open questions (subsection E), and a comparison summary with positioning grid (subsection F).

A. Method genealogy: Figure 1 shows the genealogy of Q-learning methods covered in this paper. Nodes are methods; edges are inheritance relationships labeled by *the weakness of the parent that the child method addresses*. The annotations on the edges are themselves a contribution: they trace the field’s evolution as a sequence of targeted responses rather than a chronological accumulation of techniques.

Two structural observations emerge from the genealogy. First, **Rainbow [6] is the integration point** for five separate axis responses (Double DQN, PER, Dueling, multi-step, distributional, NoisyNet) — it is not a single method but a deliberate composition across axes. Second, **the distributional family**

TABLE I
COMPARISON OF Q-LEARNING AND DEEP Q-LEARNING SURVEY PAPERS.
● = DISCUSSED; ○ = NOT DISCUSSED; PARTIAL = *PARTIAL*.

Aspect	Urtans 2018	Jang 2019	Boppiniti 2021	Hafiz 2022	Ghasemi 2024/25	Ours (2)
Analyzes Public DQN Code Repositories	○	○	○	○	○	●
Unified Taxonomy Covering Both Tabular Q and DQN	○	○	○	○	<i>partial</i>	●
Atari Benchmark Results Extracted from Prior DQN Papers	○	○	○	○	○	●
Classic Control Benchmark Results from Original Implementations	○	○	○	○	○	●
Thorough Per-Paper Literature Review	○	○	○	○	<i>partial</i>	●
Problem-First Organization Across Multiple Axes	○	○	○	○	○	●

TABLE II
THE METHOD-TYPE TAXONOMY USED BY PRIOR Q-LEARNING SURVEYS — ITS SIX CATEGORIES AND WHERE EACH CATEGORY’S METHODS FALL AMONG THIS PAPER’S EIGHT PROBLEM AXES. FAMILIES THAT POSTDATE THIS TAXONOMY (OFFLINE RL §IV.E, MULTI-AGENT §IV.F, DISTRIBUTED/META §IV.G) HAVE NO METHOD-TYPE CATEGORY AT ALL.

Method-type category	Groups methods that...	Examples	Problem axes
Statistical	model the return distribution or inject parameter noise	C51, QR-DQN, NoisyNet	§IV.C, §IV.D
Q-Function Computation	alter how the bootstrap target is computed	Double DQN, Dueling, Rainbow	§IV.A, §IV.B, §IV.H
Memory / Replay	change what experience is stored and replayed	DQN, PER, DQfD, HER	§IV.B
Ensemble-Based	maintain several Q-estimators	Bootstrapped DQN, EBQL, REDQ	§IV.A, §IV.C
Model-Based	use a model or a posterior over values	Bayesian Q, Posterior-Sampling DQN	§IV.C, §V
Pure Q-Learning	tabular or minimal value iteration	Q-learning, SARSA, NFQ	§V, §IV.H

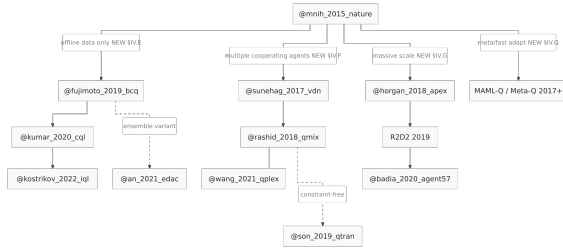


Fig. 2. Q-learning branches outside the conventional six-category mechanism taxonomy — offline RL (§IV.E), multi-agent value decomposition (§IV.F), and distributed / meta-learning (§IV.G).

(C51 → QR-DQN → IQN → FQF) constitutes an unusually long single-axis lineage compared to most other branches, reflecting the field’s progressive refinement of distributional Q-learning under the credit-assignment weakness (W4).

B. Branches outside the conventional taxonomy: Figure 2 below shows three Q-learning branches that emerged outside the conventional six-category mechanism taxonomy — families that respond to weaknesses (distribution shift, multi-agent coordination, distributed scale and meta-adaptation) that the method-type taxonomy has no category to hold. These are the methods that motivate §IV.E, §IV.F, and §IV.G.

The visual absence of these branches from prior Q-learning surveys [7–10] is one of the most direct empirical arguments for the structural pivot of §IV. The method-type taxonomy classifies methods by mechanism type; these branches require new mechanism categories (constrained policies, value decomposition, distributed actors, meta-learning) that did not exist

when the original taxonomy was conventionalized.

C. Axis × mechanism-family matrix: The matrix below summarizes which mechanism families address which axis. Filled cells indicate a primary contribution; open cells (○) indicate an incidental or secondary contribution; blanks indicate the axis is not addressed by that mechanism family.

Several patterns are visible at a glance:

- The **architectural decomposition** column is the most populous — most axes have at least one architectural response. This reflects the field’s strong preference for *structural* solutions (modifying what the network represents) over *training-procedure* solutions (modifying what the loss optimizes).
- The **distributed** column is sparse but increasingly populated by frontier agents (Ape-X, Agent57). The pattern suggests scaling-as-mechanism is undercovered in the literature relative to its empirical importance — a point the open-questions subsection of §IV.G develops.
- The **decoupling** column has only two entries (DoubleQ-derived). Decoupling is mechanistically narrow; ensembling subsumes most of its effect once compute budgets permit $K > 2$ Q-functions.
- The **bottom-left quadrant** of the matrix (W6, W7 × decoupling or ensembles) is empty. Whether this reflects a true gap in the literature or merely a labeling artifact is an open question flagged in §VIII.

D. Axis interactions and failure modes: The eight weaknesses are not independent: a method that resolves one frequently aggravates another. The table below names, for each axis, its mathematical origin, the axis it most strongly interacts with, and the deployment symptom that signals it

Axis	Decoupling	Ensembles	Architectural decomposition	Behavior / loss modulation	Distributed / parallel
W1 Overestimation (§IV.A)	DoubleQ, DDQN	EBQL, REDQ	Dueling (rel.) $\circ$	—	—
W2 Sample inefficiency (§IV.B)	—	—	—	PER, DQfD, MeDQN, HER	Ape-X (replay) $\circ$
W3 Brittle exploration (§IV.C)	—	Bootstrapped, UCB-Q	NoisyNet, ParSpaceNoise	CBDQ, PSDQN, RND, Go-Explore	Agent57 (portfolio)
W4 Reward sparsity (§IV.D)	—	—	Distributional family	n-step, λ -returns	—
W5 Distribution shift (§IV.E)	BCQ	EDAC	—	CQL, IQL, BRAC, AWAC	—
W6 Multi-agent coord. (§IV.F)	—	—	VDN, QMIX, QPLEX, QTRAN	—	—
W7 Scaling / adaptation (§IV.G)	—	—	DRQN, R2D2	—	Ape-X, Agent57, Meta-Q
W8 Function-approx stability (§IV.H)	—	—	Target net, Dueling, PQN	MeDQN consol., Munchausen	—

— a practitioner-facing complement to the mechanism view above.

E. Section roadmap: The remainder of §IV proceeds axis by axis:

- **§IV.A** Overestimation bias — Double Q-learning to ensemble bias control (EBQL, REDQ).
- **§IV.B** Sample inefficiency — prioritized replay, demonstrations, memory-efficient consolidation, hindsight relabeling.
- **§IV.C** Brittle exploration — noise injection, ensemble disagreement, belief modulation, intrinsic motivation, archival exploration.
- **§IV.D** Reward sparsity and credit assignment — multi-step returns and the distributional family.
- **§IV.E** Distribution shift (offline RL) — policy constraint, value penalty, expectile regression, ensemble diversification.
- **§IV.F** Multi-agent coordination — additive, monotonic, duplex-dueling, and constraint-free value decomposition.
- **§IV.G** Scaling and slow adaptation — distributed actor-learner architectures, bandit-controlled exploration meta-policies, meta-learning, recurrence.
- **§IV.H** Function-approximation instability — target networks, architectural decomposition, normalization recipes, consolidation losses.
- **§IV.I** Theoretical foundations and recent advances — convergence theory, finite-time bounds, pessimism in offline RL, stability theory, and IGM theorems. A meta-section that surveys the theoretical landscape underlying the eight axes above.
- **§IV.J** Q-learning for foundation model alignment — Q-Transformer, ShiQ, VLM Q-Learning, Q[‡]. An application-oriented section covering Q-learning’s role in fine-tuning large language and vision-language models, distinct from the weakness-axes in that it represents a new deployment regime rather than a new mechanism category.

Each axis subsection (A–H) ends with **F. Comparison summary**: a uniform six-column table (method / mechanism category / mechanism / cost / best-at / empirical anchor) and, where the trade-off is naturally two-dimensional, a 2D positioning grid. §IV.E additionally provides a method-selection

decision tree for practitioner use. §IV.I substitutes a results-by-year theoretical-status table for the comparison grid since its entries are theorems rather than methods. §IV.J provides a four-column comparison table organized by off-policy support and KL-regularization compatibility.

IV.A. Overestimation Bias

Weakness. The update target $r + \gamma \max_{a'} Q(s', a')$ applies a maximum over noisy estimates, which is biased upward: the expected maximum of noisy estimators exceeds the maximum of their true means [23]. Propagated through the Bellman backup the bias compounds, steering the greedy policy toward poorly-estimated actions rather than genuinely good ones — bounded in the tabular case but able to grow without bound under neural approximation.

Mechanisms. Three families debias the maximum, distinguished by *what information they exploit*. (i) *Decoupling* separates action selection from evaluation: tabular Double Q-learning [24] uses two independent estimators, and Double DQN [25] reuses DQN’s target network for evaluation at near-zero cost, $y = r + \gamma Q(s', \arg \max_{a'} Q(s', a'; \theta); \theta^-)$. A recent re-examination, DDQL [26], shows this target-network substitution is *not* equivalent to true two-estimator decoupling — it carries a stale-bootstrap bias — and training two independent networks restores lower bias and improves on 47 of 57 Atari games under matched compute. (ii) *Ensemble* methods generalize to K estimators and make the bias–variance trade-off tunable: EBQL [27] evaluates with the average of held-out members ($K \in [5, 10]$ interpolating between DQN’s over- and Double DQN’s under-estimation), while REDQ [28] takes a minimum over a random subset to deliberately induce under-estimation. (iii) *Architectural* decomposition — Dueling DQN [29] — routes learning through a state-value baseline $V(s)$, reducing the relative noise in the action-conditional term; this is primarily a stability contribution (§IV.H)(#sec-iv-h)) that incidentally dampens the max bias. For *hybrid* discrete–continuous actions, PDQN [30] debiases the discrete max while a learned actor emits the continuous parameters, avoiding bias compounding across the two levels.

Trade-off. No fix dominates. Double DQN trades a small under-estimation for removing a large over-estimation at essentially no cost; ensembles buy *calibratable* bias at $K \times$ compute

TABLE III

EACH WEAKNESS, ITS ORIGIN, THE AXIS IT MOST STRONGLY INTERACTS WITH, AND THE SYMPTOM THAT SIGNALS IT IN DEPLOYMENT. MOST INTERACTION PATHS LEAD BACK TO W8 (FUNCTION-APPROXIMATION STABILITY), WHICH THE MATRIX ABOVE ALREADY SHOWS AS THE FIELD'S CENTRAL FAILURE MODE.

Axis	Mathematical origin	Principal interaction	Deployment failure mode
W1 Overestimation	max over noisy estimates biased upward	debiasing can suppress optimism (W3)	value divergence in stochastic / large action spaces
W2 Sample inefficiency	uniform replay ignores information density	heavy reuse of stale data erodes stability (W8)	slow convergence under memory limits
W3 Brittle exploration	temporally inconsistent ϵ -greedy dithering	poor coverage delays credit propagation (W4)	stuck on sparse-reward plateaus (e.g. Montezuma)
W4 Credit assignment	one-step backups propagate reward slowly	n -step cuts bias but adds off-policy variance (W8)	fails under delayed / terminal-only rewards
W5 Distribution shift	bootstrapping on OOD actions, narrow support	OOD extrapolation mimics overestimation (W1)	offline policy collapse without interaction
W6 Multi-agent	non-stationarity breaks the single-agent MDP	exponential joint action space strains stability (W8)	coordination collapse under decentralized execution
W7 Scaling / adaptation	sequential-interaction bottleneck; no meta-transfer	stale actor-learner gradients destabilize targets (W8)	extreme sample / wall-clock cost; weak online adaptation
W8 Instability	deadly triad: off-policy + bootstrap + approximation	target nets stabilize but slow learning (W2)	catastrophic value divergence, oscillation

TABLE IV

OVERESTIMATION-BIAS METHODS (SEE SECTION V FOR BENCHMARK PROTOCOL CAVEATS).

Method	Mechanism	Cost	Best at
Double Q (2010)	Two swapped estimators	None	Tabular MDPs
Double DQN (2016)	Online selects, target evaluates	Free	Most Atari
DDQL (2026)	Two independent nets	$2\times$	47/57 Atari
EBQL (2021)	Avg. of $K-1$ held-out	$K\times$	Bias calibration
REDQ (2021)	Min of M random members	$K\times$	Continuous control
Dueling* (2016)	$V(s) + A(s, a)$ centered	Modest	Many-action states

*Primary axis is stability (Section IV-H); listed here as an incidental contributor.

and memory; REDQ's minimum helps when the optimizer drifts toward high- Q regions but hurts when optimism is needed, as in long-horizon tasks that depend on exploration. The right operating point therefore depends on action-space geometry and on whether the downstream policy is more sensitive to over- or under-estimation. Critically, these *online* fixes transfer imperfectly to offline RL, where overestimation re-appears as out-of-distribution action selection and demands different tools (CQL, IQL; [§IV.E](#sec-iv-e)).

Open questions. The bias-variance Pareto frontier as a function of ensemble size K , subset size M , and update rate is unmapped; and the relationship between online debiasing and offline OOD-penalization remains ad hoc, lacking a framework that recovers both as limits of a single inequality.

IV.B. Sample Inefficiency

Weakness. Vanilla Q-learning consumes each transition once; experience replay [18] relaxes this by sampling minibatches from a buffer $\mathcal{D}$, $\theta \leftarrow \theta - \alpha \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}} [\nabla_{\theta} (Q(s,a;\theta) - y)^2]$. But uniform sampling treats every transition as equally informative, so when high-information transitions are rare — sparse rewards,

low-probability states, narrow-margin decisions — learning stays bottlenecked on the rate at which those transitions reappear in the sample.

Mechanisms. Four families recur, distinguished by *what they manipulate*. (i) *Prioritized sampling*: Prioritized Experience Replay (PER) [31] draws transitions in proportion to TD-error magnitude, $P(i) \propto (|\delta_i| + \epsilon)^\alpha$, with importance-sampling weights annealed toward 1 to correct the induced bias; it is among the two largest contributors to Rainbow [6]. (ii) *Demonstration augmentation*: Deep Q-learning from Demonstrations (DQfD) [32] seeds the buffer with expert trajectories and adds a large-margin supervised term, trading exploration cost for demonstration cost to produce far stronger initial policies. (iii) *Memory-efficient consolidation*: MeDQN [33] replaces most of the buffer with a consolidation loss that distills past Q -values from the target network, cutting Atari storage roughly tenfold; the same mechanism damps function-approximation drift and so doubles as a stability tool ([§IV.H](#sec-iv-h)). (iv) *Goal relabeling*: Hindsight Experience Replay (HER) [34] rewrites failed goal-conditioned trajectories as successes under synthetic goals $g' = s_T$, densifying reward at no environmental cost.

Trade-off. No method dominates, because the relevant axes are categorical rather than a clean bias-variance frontier. PER's prioritization is asymptotically unbiased but injects variance early in training, tuned through α and the β schedule. DQfD's gain scales with demonstration quality and is inapplicable when expert data is absent or reward-misaligned. MeDQN trades buffer storage for the compute of its consolidation loss plus a weight λ to tune per domain. HER applies only to MDPs whose reward decomposes over goals, leaving non-goal-conditioned sparsity untouched. The choice therefore turns on environment structure and resource budget, not a single dominant operating point. Crucially, these gains are tied to the *online* replay distribution: PER's correction is computed against the current buffer, so its transfer to the fixed distribution of offline RL ([§IV.E](#sec-iv-e)) is unsettled and empirically mixed.

Open questions. FIFO replacement is the unjustified default

and discards the rare early successes of highest information density, leaving adaptive, utility-aware replacement largely unexplored. And goal relabeling beyond Cartesian goals — rewards computable only from full trajectory features — lacks a parameterizable relabeling space.

IV.C. Brittle Exploration

Weakness. ϵ -greedy takes a uniform-random action with probability ϵ and otherwise acts greedily — *dithered* noise applied independently per step, with no memory of past exploration and no model of which actions remain uncertain. It is myopic with respect to epistemic uncertainty: a state-action pair visited many times is sampled at the same rate as one visited never, so it fails when reward is sparse, delayed, or behind passages whose random-walk hitting time is exponential in trajectory length — the canonical case being Montezuma’s Revenge [2], where DQN scores essentially zero.

Mechanisms. Four families induce a posterior over $Q(s, a)$ or an exploration bonus, distinguished by *what they exploit*. (i) *Noise injection* perturbs the network rather than the action: parameter-space noise [35] adds $\mathcal{N}(0, \sigma^2 I)$ to θ once per episode (with layer normalization [36] and adaptive σ) for temporally consistent exploration, while NoisyNet [37] replaces weights with learned $\mu + \sigma \odot \epsilon$ sampled per forward pass so the network *learns* where to keep exploring — a load-bearing component of Rainbow [6]. (ii) *Ensemble disagreement* treats variance across K Q-heads as epistemic signal: Bootstrapped DQN [38] samples one bootstrap-trained head per episode, and UCB Q-Ensembles [39] act on $\arg \max_a (\mu(s, a) + \lambda \sigma(s, a))$ — the same ensemble that serves bias control in [§IV.A](#sec-iv-a). (iii) *Belief / posterior modulation* reweights backups: CBDQ [40] maintains a clustered belief $b_t(a | s)$ that weights future values, while Posterior Sampling DQN [41] draws a Q -hypothesis from an approximate posterior [42] per episode (deep Thompson sampling). (iv) *Intrinsic motivation* modifies reward: RND [43] adds a bonus from random-network prediction error that decays as states become familiar, the first method to reach non-trivial Montezuma scores; Go-Explore [44] instead archives visited states and *returns-then-explores*, decoupling reaching a state from exploring it, and fully solves Montezuma’s Revenge.

Trade-off. No method dominates. Per-episode perturbation (parameter noise, Bootstrapped DQN) yields multi-step consistency that solves partially-observed tasks like flickering Pong [45], whereas NoisyNet’s per-pass sampling adapts faster but loses that property. Ensembles multiply forward-pass cost and memory by K (largely hidden by batching). Intrinsic bonuses add a scale hyperparameter and risk novelty-addiction that crowds out extrinsic reward. Go-Explore is empirically dominant on hard-exploration Atari but introduces a non-Markovian archive whose extension to continuous spaces is unsettled. Crucially, demonstrations remain the most effective sidestep on the hardest games — Rainbow’s best honest Montezuma result trails DQfD by an order of magnitude — sharpening the exploration-vs-demonstration tension of [§IV.B](#sec-iv-b); distributional gains ([§IV.D](#sec-iv-d)) do not transfer to this axis at all.

Open questions. All methods here fix one exploration strategy for all of training, yet game-by-game variance argues for adaptive selection (meta-learned, bandit-controlled as in Agent57 [46] [§IV.G](#sec-iv-g), or uncertainty-thresholded), and no consensus exists on balancing intrinsic against extrinsic reward or on archive-based exploration in continuous latent spaces.

IV.D. Reward Sparsity and Credit Assignment

Weakness. The one-step Q-learning update propagates reward by exactly one Bellman backup per environment step, so when reward arrives only at the end of a long trajectory the signal must traverse the whole horizon through repeated updates before it reaches early states. This is the credit-assignment problem: apportioning fractional responsibility for a terminal reward across upstream state-action pairs, which a scalar expected-return target conveys slowly and with little structure.

Mechanisms. Two families attack the weakness by changing *what the return signal carries*. (i) *Temporal-extent* methods widen the backup: Multi-Step Q-Learning [19] uses n -step bootstrapped returns with eligibility traces, $y_t^{(n)} = r_t + \gamma r_{t+1} + \dots + \gamma^{n-1} r_{t+n-1} + \gamma^n \max_{a'} Q(s_{t+n}, a')$, and the TD(λ) interpolation tunes bias against variance via λ ; in Rainbow [6] three-step returns are the second-largest performance contributor after prioritized replay. (ii) *Information-content* methods replace the scalar $\mathbb{E}[\text{return}]$ with the full return distribution $Z(s, a)$. C51 [20] models Z as a categorical distribution over 51 fixed atoms on a bounded support $[V_{\min}, V_{\max}]$ and minimizes a projected-KL target; QR-DQN [47] inverts this with fixed probabilities and *learned* quantile locations under a quantile Huber loss, removing the projection and the support prior; IQN [48] samples quantile fractions $\tau \sim \mathcal{U}(0, 1)$ at runtime to learn a continuous distribution, reaching Rainbow-comparable scores alone; and FQF [49] additionally *learns* the fractions per state-action pair, attaining the highest median Atari at roughly 20% more compute than IQN.

Trade-off. No method dominates. Larger n or λ propagates reward faster but inflates variance and off-policy bias, so three-step is the practical sweet spot for single-stream agents while distributed throughput (Ape-X, R2D2) makes longer horizons viable. Across the distributional family, resolution trades against compute along the monotone-in-median diagonal $\text{C51} \rightarrow \text{QR-DQN} \rightarrow \text{IQN} \rightarrow \text{FQF}$, yet per-game behavior is non-monotone. The quantile-regression methods also incur a *quantile-crossing* pathology: training independent estimators per τ_i without enforcing $\tau_i < \tau_j \Rightarrow \theta_i \leq \theta_j$ yields non-monotone pseudo-distributions, worst early in training and on asymmetric rewards; non-crossing variants smooth the curves at small median gains. Interpretively, this paper reframes the distributional family as a *credit-assignment* mechanism rather than the conventional uncertainty grouping ([§IV.G](#sec-iv-g)): all these methods still *act* on $\mathbb{E}[Z(s, a)]$, so the richer distribution helps *learning*, not *acting*, and its strength on long-horizon tasks reflects better backup propagation, not optimism over uncertainty. The flip side is that richer signal cannot substitute for directed search — on the hardest explo-

TABLE V
SAMPLE-EFFICIENCY METHODS.

Method (year)	Mechanism	Cost	Best at
DQN replay (2013)	Uniform sampling from buffer	Memory	Off-policy baseline
PER (2016)	Sample $\propto \delta_i ^\alpha$, IS-corrected	IS bias, α tuning	Rare high-info transitions
DQfD (2018)	Buffer seeded with expert demos	Demonstration availability	Sparse-reward Atari
MedQD (2023)	Consolidation loss compresses buffer	λ hyperparameter	Memory-constrained training
HER (2017)	Goal relabeling for synthetic reward	Goal-conditioned only	Robotic manipulation

TABLE VI
EXPLORATION METHODS.

Method (year)	Mechanism	Cost	Best at
Param Space Noise (2017)	Per-episode θ perturbation	Layer norm	Episode-consistent exploration
NoisyNet (2018)	Per-pass weight noise, learned σ	$2\times$ forward	Adaptive exploration
Bootstrapped DQN (2016)	K -head ensemble, head per episode	$K\times$ memory	Episode-coherent exploration
UCB Q-Ensemble (2018)	$\mu + \lambda\sigma$ over K -ensemble	$K\times$ compute	Uncertainty-aware action
CBDQ (2025)	Belief over actions weights backups	Clustering	Classic control + driving
Posterior Sampling DQN (2023)	Sample Q from posterior per episode	Approx. posterior	Cyclic environments
RND (2018)	Random-network distillation bonus	Bonus scale	Sparse-reward Atari
Go-Explore (2019/21)	Archive + return-then-explore	State hashing	Hardest Atari exploration

ration games ([§IV.C](#sec-iv-c)) the family scores near zero, sharpening the W3/W4 separation.

Open questions. Convergence guarantees for the quantile family under function approximation remain substantially weaker than for C51’s projection step, and whether its compositional gain with prioritized replay is orthogonal or partly redundant is unsettled given IQN’s standalone strength. Whether these architectures support stable risk-sensitive action selection on tail behavior, rather than acting on the mean, is largely unexplored.

IV.E. Distribution Shift (Offline Reinforcement Learning)

Weakness. Q-learning’s off-policy property is theoretical: the Bellman operator is independent of the data-generating policy, but in *offline* RL the buffer is fixed and the bootstrap target $r + \gamma \max_{a'} Q(s', a')$ ranges over the entire action space, including out-of-distribution (OOD) actions no behavior policy π_b ever took. The approximator extrapolates arbitrarily at such points, and iterated backups drive Q unboundedly upward where data support vanishes — so naive offline Q-learning consistently underperforms even the behavior policy [50].

Mechanisms. Four families respond, distinguished by *what they constrain*. (i) *Policy constraint* restricts the learned policy toward π_b : BCQ [50] samples actions from a learned behavior model plus small perturbations; BRAC [51] adds an explicit KL/Wasserstein divergence penalty; and AWAC [52] uses an advantage-weighted update $\pi(a | s) \propto \pi_b(a | s) \exp(A(s, a)/\beta)$ that bridges to online fine-tuning. (ii) *Value penalty* — CQL [53] adds a regularizer that pushes Q down at OOD actions and up on in-distribution ones, yielding a lower bound on V^π . (iii) *Avoiding the max* — IQL [?] drops the max entirely, learning a state value via expectile regression and backing up $Q(s, a) \leftarrow r + \gamma V(s')$, so extrapolation error is structurally prevented. (iv) *Ensemble diversification* — EDAC [54] takes a min over K critics trained to disagree

on OOD actions; Cal-QL [55] calibrates CQL’s bounds for the offline-to-online bridge. A newer line couples expressive generative policies to Q-learning: FQL [56] trains a one-step flow-matching policy, sidestepping backpropagation through a diffusion chain, while adjoint-matching variants tolerate full multi-step generation and recover the optimal behavior-regularized policy at convergence.

Trade-off. No method dominates, and the axis trades cleanly against itself. Policy-constraint methods are safe but cannot outperform the best in-support trajectory by much; CQL’s penalty weight α is its single most sensitive hyperparameter and varies across tasks despite adaptive scaling [57]; IQL avoids extrapolation but weakens formal optimality guarantees; EDAC’s K -network ensemble multiplies compute and needs per-action gradients; flow policies trade expressiveness against tractability. The deeper connection is to [§IV.A](#sec-iv-a): offline OOD action selection *is* overestimation bias under a fixed buffer — the online max-over-noisy-estimates problem re-appearing where the correction signal of fresh on-policy samples is absent. The online debiasing tools transfer imperfectly, which is precisely why the offline setting demands its own conservatism (CQL, IQL) rather than the decoupling and ensembles that suffice online.

Open questions. A unifying framework recovering all four families as limits of a single regularization remains absent, and whether Q-learning’s mechanism suits internet-scale heterogeneous data — where sequence-modeling approaches currently dominate — is empirically open with substantial practical stakes.

IV.F. Multi-Agent Coordination

Weakness. Cooperative multi-agent RL models N agents in a shared-reward dec-POMDP where each agent acts on a local observation-action history τ_i but the reward depends on the joint action. Independent Q-learning treats co-learners as

TABLE VII
CREDIT-ASSIGNMENT AND DISTRIBUTIONAL METHODS.

Method (year)	Mechanism	Cost	Best at
Multi-Step Q (1996)	n -step returns + eligibility traces	Variance under off-policy	Long-horizon credit assignment
C51 (2017)	51 fixed atoms, projected KL	Bounded support pre-specified	Atari mean + median
QR-DQN (2018)	N learned quantile locations	Quantile loss, no projection	Dense + strategic Atari
IQN (2018)	Runtime-sampled $\tau \sim U(0, 1)$	Cosine fraction embedding	Standalone Rainbow-comparable
FQF (2019)	Learned quantile fractions + values	+20% compute vs. IQN	Highest median Atari

TABLE VIII
OFFLINE Q-LEARNING METHODS.

Method (year)	Mechanism	Cost	Best at
BCQ (2019)	Generative behavior model + perturbation	Generator quality	Narrow-support data
BRAC (2019)	KL/Wasserstein divergence penalty	Divergence weight tuning	Diverse offline data
AWAC (2020)	Advantage-weighted offline $\rightarrow$ online	β tuning	Offline-to-online bridge
CQL (2020)	Penalize Q at OOD actions	α sensitivity	Random/medium data
IQL (2021)	Expectile V , Q backed up via $V(s')$	τ tuning	Single-network simplicity
EDAC (2021)	Min over K -ensemble + gradient diversity	$K \times$ compute	Locomotion strongest
Cal-QL (2023)	Calibrated conservative bounds	α tuning	Offline-to-online bridge
FQL (2025)	One-step flow-matching policy + Q	Flow training	Expressive offline policies

non-stationary dynamics and destabilizes, while a centralized joint Q_{tot} over the product action space scales exponentially in N . The crux is the **Individual-Global-Max (IGM)** property: a factorization $Q_{\text{tot}} = f(Q_1, \dots, Q_N)$ must let each agent act greedily on its local Q_i yet recover the joint greedy action, $\arg \max_{\mathbf{a}} Q_{\text{tot}} = (\arg \max_{a_1} Q_1, \dots, \arg \max_{a_N} Q_N)$, so that decentralized execution stays globally optimal.

Mechanisms. Decomposition families trade representational capacity against training stability by how strictly they constrain f . (i) *Additive* — VDN [58] sets $Q_{\text{tot}} = \sum_i Q_i(\tau_i, a_i)$, trivially IGM-satisfying and trivial to train, but unable to express non-separable coordination. (ii) *Monotonic* — QMIX [59] relaxes additivity to $\partial Q_{\text{tot}} / \partial Q_i \geq 0$ via a state-conditioned hypernetwork mixer, preserving IGM with richer capacity and becoming the dominant baseline. (iii) *Full-expressiveness via dueling* — QPLEX [60] mixes per-agent advantage streams under an IGM-preserving constraint, representing the entire IGM class. (iv) *Constraint-free* — QTRAN [61] learns an unconstrained joint Q and enforces IGM through auxiliary losses, maximal in principle but hard to optimize.

Trade-off. The progression $\text{VDN} \rightarrow \text{QMIX} \rightarrow \text{QPLEX} \rightarrow \text{QTRAN}$ raises representational capacity monotonically, but capacity and stability are *not* independent: the empirical SMAC ordering is $\text{QMIX} \approx \text{QPLEX} > \text{VDN} > \text{QTRAN}$, because structural constraints that buy expressiveness also degrade training dynamics. QPLEX’s full expressiveness pays off only on super-hard scenarios where representation is the bottleneck, whereas QTRAN’s theoretical maximum never realizes — its loss-enforced IGM is the canonical case of capacity without trainability. QFIX/Q+FIX [62] reframes this tension as a *base-plus-correction* pattern rather than a monolithic mixer redesign: a small per-agent residual network corrects a VDN, QMIX, or QPLEX base toward the full IGM class, recovering the QPLEX expressiveness ceiling with simpler

training dynamics and consistent gains over both QMIX and QPLEX on SMACv2 and Overcooked. This decouples the two axes that the earlier methods conflated. These instabilities compound the single-agent biases of [§IV.A](#sec-iv-a) and [§IV.H](#sec-iv-h), and the required centralized critic limits the framework to settings where centralized training is feasible.

Open questions. The IGM frame assumes a single optimal joint policy and greedy decentralized execution, leaving equilibrium selection, communication, and heterogeneous agents to ad-hoc extensions. Multi-agent off-policy correction [63] and the offline multi-agent regime — where per-agent behavior constraints need not yield coordinated joint behavior — remain among the cleanest cross-axis open problems.

IV.G. Scaling and Slow Adaptation

Weakness. This axis is *composite*: vanilla Q-learning is both sample-throughput-bound (W7a) and slow to adapt across tasks (W7b). A single sequential learner cannot consume transitions faster than the environment produces them — collecting Rainbow’s [6] 200M Atari frames takes weeks of wall-clock time — while a Q-function trained on task $\mathcal{T}_1$ specializes to $\mathcal{T}_1$ and neither retraining nor warm-starting from $Q^{\mathcal{T}_1}$ yields few-shot transfer to a related $\mathcal{T}_2$. The two failure modes are bundled because the methods that answer either mostly answer both.

Mechanisms. The W7a (throughput) cluster parallelizes interaction. Ape-X [64] decouples collection from learning — many actors feed a centralized prioritized buffer that one GPU learner drains — for $\sim 50\times$ wall-clock speedup; R2D2 [65] adds an LSTM agent with replay *burn-in* to re-sync stale recurrent states, building on DRQN’s [45] recurrent head for partial observability; Agent57 [46] adds an NGU intrinsic-motivation module and a bandit over an exploration-discount

TABLE IX
VALUE-DECOMPOSITION METHODS.

Method (year)	Representational capacity	Training stability	Core mechanism
VDN (2018)	Lowest (additive only)	Highest	$Q_{\text{tot}} = \sum_i Q_i$
QMIX (2018)	Medium (monotonic)	High	State-conditioned monotonic hypernetwork mixer
QPLEX (2020)	High (full IGM class)	Medium	Duplex dueling on advantage streams
QTRAN (2019)	Maximum (unconstrained)	Lowest	Unconstrained joint Q + auxiliary IGM losses
QFIX/Q+FIX (2025)	High (recovers IGM ceiling)	High	Per-agent residual correction on a base mixer

policy portfolio, the first to clear human level on all 57 games. PQN [66] instead uses synchronous vectorized environments, trading actor-learner staleness for single-machine simplicity ([§IV.H](#sec-iv-h)). The W7b (adaptation) cluster meta-trains across a task distribution $p(\mathcal{T})$, differing in *what is meta-learned*: an initialization (MAML-Q [67, 68], the first-order Reptile-Q [69], proximal ProMP [70])

$$\theta_0^* = \arg \min_{\theta_0} \mathbb{E}_{\mathcal{T}} [\mathcal{L}_{\mathcal{T}}(\theta_0 - \alpha \nabla_{\theta_0} \mathcal{L}_{\mathcal{T}}(\theta_0))];$$

a context embedding z that conditions $Q(s, a, z)$ with no gradient adaptation (PEARL [71], off-policy MQL [72]); or a forward-pass update rule, where a transformer reads the recent trajectory and predicts actions in-context — algorithm distillation and AdA, consolidated since 2023 into in-context Q-learning proper (SICQL [73], compositional ICQL [74]).

Trade-off. The deep distinction is between *hardware-driven* throughput and *algorithmic* few-shot adaptation. W7a methods buy performance with parallelism and compute: the Ape-X→R2D2→Agent57 progression is the single largest source of Atari gains since DQN, suggesting throughput is the binding constraint at scale and that algorithmic axes become secondary once compute is abundant. W7b methods instead buy data-efficiency on new tasks — adapting in 5–50 episodes where single-task baselines need thousands — at 10–100× meta-training cost whose payoff is empirically mixed. Methods span both because the distributed compute W7a developed is precisely what makes meta-training and large in-context models feasible, and because *implicit* meta-learning (a single Agent57/AdA-style agent trained across the whole distribution) often matches *explicit* outer-loop meta-learners while also exploiting the distributed actor pool. Predictable scaling laws [75] reframe both: the data-versus-compute Pareto follows the updates-to-data ratio, so the optimal allocation can be extrapolated from a small pilot — orthogonal to, and compatible with, the architecture that supplies the throughput.

Open questions. Whether meta-Q methods extend beyond narrow parametric task distributions to heterogeneous state/action/reward families is unresolved, and no compute-matched, axis-stratified ablation exists to separate genuine algorithmic gains from gains that are merely compensating for scale.

IV.H. Function-Approximation Instability

Weakness. The *deadly triad* [21] names the conjunction of off-policy learning, bootstrapping, and function approx-

imation: any two are stable together, but all three admit counterexamples in which the semi-gradient update

$$\theta \leftarrow \theta + \alpha (r + \gamma \max_{a'} Q(s', a'; \theta) - Q(s, a; \theta)) \nabla_{\theta} Q(s, a; \theta)$$

diverges or oscillates. The instability arises because each gradient step also moves Q at the bootstrap target $Q(s', a')$, making the update self-referential and potentially expansive — deep Q-learning trained naively does diverge.

Mechanisms. Four families damp the triad, distinguished by *what they exploit*. (i) *Target decoupling* freezes a copy $Q(\cdot; \theta^-)$ for a window so that $y = r + \gamma \max_{a'} Q(s', a'; \theta^-)$ no longer chases its own parameters [1]; this is the single load-bearing stabilizer of the classical recipe, whether updated by hard copy or Polyak averaging. (ii) *Architectural decomposition* — Dueling [29] — routes most of the gradient through a state-value baseline, reducing noise in the action-conditional term; we re-interpret its contribution as primarily stability, consistent with the Rainbow ablation showing it the smallest-impact component [6] (its incidental bias effect appears in [§IV.A](#sec-iv-a)). (iii) *Normalization* damps the expansive map directly: PQN [66] applies LayerNorm [36] throughout the body, input BatchNorm, optional L2, and n -step returns, while Munchausen DQN adds a $\log \pi$ reward bonus that implicitly regularizes toward the previous iterate. (iv) *Consolidation* — MeDQN [33], kin to elastic weight consolidation [76] — penalizes movement away from past Q -values to counter the non-stationarity that drives drift.

Trade-off. No method dominates, because the recipes trade staleness, sample efficiency, and complexity against one another. Hard target updates buy stability at the cost of targets up to C steps stale; soft updates smooth that staleness but slow responsiveness; consolidation and Munchausen bonuses add a hyperparameter (λ, τ) that under- or over-damps. The pivotal result is PQN’s: target networks *and* experience replay — long held essential — can be removed entirely, with LayerNorm plus parallel vectorized environments matching PER [31], Double DQN [25], and Rainbow [6] on Atari at up to 50× less wall-clock. The triad’s instability is thus dampable by normalization alone, the classical stabilizers being sufficient but not necessary. The catch is that PQN substitutes parallel environments for replay’s sample reuse, so where interaction is expensive (real robotics) the trade-off may still favor replay retention.

Open questions. A formal stability guarantee for normalized Q-learning under the deadly triad — beyond informal Lip-

TABLE X
DISTRIBUTED-SCALING AND ADAPTATION METHODS.

Method (year)	Sub-axis (W7a/W7b)	Mechanism	Best at
DRQN (2015)	W7b	LSTM head over DQN	Partial observability
Ape-X (2018)	W7a	Async actors + central learner + PER	50x wall-clock speedup
R2D2 (2019)	W7a	Ape-X + LSTM + replay burn-in	Memory-demanding tasks
Agent57 (2020)	W7a/W7b	R2D2 + NGU + bandit policy portfolio	All 57 Atari at human level
PQN (2024)	W7a	Synchronous vectorized envs + LayerNorm	Compute-efficient single machine
MAML-Q / Reptile-Q / ProMP (2017+)	W7b	Meta-learn fast-adapt initialization	Few-shot gradient transfer
PEARL / MQL (2019/20)	W7b	Inferred task-context embedding	Adaptation without gradient steps
SICQL / ICQL (2026)	W7b	In-context Q via transformer forward pass	Compositional task transfer

schitz arguments — remains absent, as does a direct test isolating Dueling’s decomposition on vanilla DQN. Whether online normalization recipes transfer to offline regimes ([§IV.E](#sec-iv-e)), where extrapolation to unsupported actions is not damped by target networks, is largely unexplored.

IV.I. Theoretical Foundations and Recent Advances

Classical theory characterizes Q-learning in regimes today’s methods have outgrown. Tabular Q-learning converges with probability one to Q^* under standard stochastic-approximation conditions [Watkins & Dayan 1992], the proof reducing the Bellman optimality operator to an ℓ_∞ contraction. Once off-policy sampling, bootstrapping, and function approximation combine — the deadly triad of [§IV.H](#sec-iv-h) — iterates can diverge even when an optimal linear fit exists [Baird 1995; Tsitsiklis & Van Roy 1997], and no fixed-point theorem covers the general combined operator; the GTD/TDC family recovers off-policy linear convergence by minimizing an explicit MSPBE objective [77].

Recent work targets the neural and offline regimes where modern methods actually run. Offline RL is the most mature axis: pessimistic value iteration attains $\tilde{O}(\sqrt{1/n})$ suboptimality without behavior-coverage assumptions [Jin, Yang & Wang 2021], grounding the CQL/IQL penalties of [§IV.E](#sec-iv-e), with extensions to multi-task settings [78]. Finite-time bounds for neural fitted-Q hold only under restrictive complexity and mixing assumptions [79]. Stability theory is catching up to practice: capacity loss [80] and the broader plasticity-loss literature [Klein et al. 2026] now formalize why deep Q-networks degrade. Distributional methods enjoy clean tabular accounts — the distributional Bellman operator is a γ -contraction in Wasserstein- p [20], and quantile regression in Wasserstein- ∞ [81] — but neural extensions remain open, as does a complete IGM characterization for multi-agent decomposition [Wang et al. 2020]. The recurring pattern is that empirical advance precedes theory by years.

Formal statements and proof sketches are provided in the supplementary material.

IV.J. Q-Learning for Foundation Model Alignment

Aligning large language and vision-language models recasts a familiar problem in unfamiliar dimensions, and that change

of scale is what turns it into a Q-learning problem. Each generated token is an action drawn from a vocabulary of 10^4 – 10^5 entries, so a single response is a long trajectory through an enormous discrete action space. The reward is sequence-level and sparse: an entire generation receives one scalar from a learned reward model, with no built-in credit assignment to the tokens that earned it — a sparse-reward problem ([§IV.D](#sec-iv-d)) at extreme scale. Finally, alignment must not destroy the pretrained model’s fluency, so the objective is regularized by a KL penalty against a reference policy, replacing the usual return with a return-minus-KL target:

$$Q_{\text{reg}}(s, a) = r(s, a) + \gamma \mathbb{E}[V_{\text{reg}}(s')] - \beta \log \pi_{\text{ref}}(a | s).$$

Methods cluster by how they exploit this structure. Offline trajectory and value methods reuse fixed data, including failed attempts: Q-Transformer [82] trains an autoregressive transformer Q-network over discretized action tokens with conservative penalties ([§IV.E](#sec-iv-e)), and VLM Q-Learning carries the same off-policy recipe to multimodal (image, text, action) data. A second group reinterprets the model’s own machinery: ShiQ treats per-token logits as Q-values up to a bias correction, making temporal-difference learning directly compatible with the KL-regularized objective and removing the online rollouts that policy-gradient RLHF requires. A third, Q-sharp (Q), targets provable behavior by guiding the reference policy with a *distributional* Q-function, trading lower KL for a given reward gain. In-context variants — SICQL and ICQL — fit transformer Q-functions that adapt at inference time with no gradient updates, and reward-shaping approaches such as Q-shaping inject value estimates as auxiliary shaping signal rather than as the optimization target.

This is frontier, fast-moving work: the benchmarks are immature (reward models judging reward models), the methods are months apart, and value-based alignment may eventually fold into a broader axis on reward-model robustness. Its long-term place in the taxonomy is not yet settled, but its present importance to the field is.

V. ATARI BENCHMARK ANALYSIS

We read the historical Atari record diagnostically rather than as a leaderboard. Games are stratified by task category

TABLE XI
STABILITY METHODS.

Method (year)	Mechanism	Cost	Best at
Target Network (2015) Dueling DQN (2016)*	Frozen Q copy for bootstrap target $V(s) + \text{centered } A(s, a)$ decomposition	Target staleness (C steps) Modest	Foundational stability Many-action states
Munchausen DQN (2020) MeDQN (2023) PQN (2024)	$\log \pi$ reward bonus, implicit KL Past- Q consolidation loss LayerNorm + n -step + parallel envs; no target net	τ hyperparameter λ hyperparameter Compute-structure shift	Implicit regularization Catastrophic forgetting Modern norm-only recipe

— reaction-time / dense-reward, strategic-planning, sparse-reward, and large-observation — and each category preferentially stresses one or two of the weakness axes of §IV: reaction-time and large-observation games load function-approximation stability (W8), strategic-planning games load credit assignment (W4), and sparse-reward games load brittle exploration (W3). The question is therefore not which method scores highest, but whether methods targeting a given weakness excel on its diagnostic category and sit near baseline elsewhere — the central empirical prediction of the taxonomy. Scores are extracted from the original publications under each paper’s own protocol; the full per-game tables (supplementary material) report the raw numbers, and we draw only category-level patterns from them here. The aim is to test predictions, not to crown a method, and where a pattern could equally be explained by reporting conventions we say so.

Sparse-reward games discriminate exploration, where the axis matters most. On Montezuma’s Revenge, Pitfall!, and Private Eye, methods that target exploration explicitly — NoisyNet, Bootstrapped DQN, posterior sampling — produce nonzero Montezuma scores where vanilla DQN and methods aimed at other axes do not, and Rainbow’s small Montezuma score is attributable to its NoisyNet component. Demonstration-based DQfD scores far higher still, illustrating the trade-off of §IV.B)(#sec-iv-b): demonstrations substitute for directed exploration. Crucially, intrinsic motivation is necessary but not sufficient — the category is not saturated by the methods tabulated here, since RND and Go-Explore (§IV.C)(#sec-iv-c)) report Montezuma scores one to two orders of magnitude larger again.

Strategic-planning games favor distributional methods. On credit-assignment-heavy games such as *Qbert* and *Ms. Pac-Man*, *quantile and distributional methods (QR-DQN, IQN, FQF)* produce the category’s most striking results — QR-DQN’s *Qbert* score is the single largest in the compilation, far above the best non-distributional result on that game — consistent with modeling the full return distribution — rather than only its mean — helping propagate value over the long action chains these games reward (§IV.D)(#sec-iv-d)). The magnitude of the gap warrants caution: such extreme scores can reflect a handful of favorable seeds as much as a mechanism, so we treat the *consistency* of the distributional advantage across the category, not any single number, as the signal. Prioritized replay shows the same category profile, strongest exactly where high-information transitions are rare and uniform sampling wastes updates.

Dense-reward and large-observation games show that stability, not capability, is the bottleneck. On reaction-time control (Breakout, Space Invaders, Enduro) and large-observation games (River Raid, Hero), inter-method variance is comparatively small: Double DQN, Dueling DQN, Rainbow, and distributional methods cluster within roughly a factor of two. Here overestimation control buys only modest gains over the DQN baseline: when rewards are frequent the agent receives enough signal that the binding constraint is keeping training stable rather than extracting value from rare events. The weakness signal is therefore function-approximation stability (§IV.H)(#sec-iv-h)) rather than capability, and PQN’s strong showing — from a deliberately simple parallelized Q-learner — is consistent with that reading: where stability is the issue, architectural restraint suffices and elaborate debiasing or distributional machinery adds little.

These patterns are suggestive, not conclusive. The original-paper numbers are point estimates over three to five seeds without confidence intervals, gathered under heterogeneous no-op-start, sticky-action, and frameskip protocols, so apparent gaps may reflect protocol or compute differences rather than algorithmic superiority. [83] document how this standard practice systematically misrepresents performance, and introduce the *reliable* framework, which substitutes robust aggregate metrics (interquartile mean, optimality gap, probability of improvement) and stratified bootstrap intervals for point-estimate comparisons. The deterministic-by-default Arcade Learning Environment [2, 84] compounds the concern, since methods exploiting frame-perfect sequences can score artificially well. We therefore refrain from declaring per-game winners and read the category narrative as the load-bearing evidence.

Atari also leaves capabilities undiagnosed, motivating newer benchmarks: ProcGen [85] measures generalization across procedural variation, NetHack [86] tests exploration and credit assignment at extreme horizons, BSuite [87] isolates single capabilities in an explicitly axis-stratified manner, D4RL [88] diagnoses offline distribution shift (§IV.E)(#sec-iv-e)), and SMAC [89] targets multi-agent coordination (§IV.F)(#sec-iv-f)).

VI. EMPIRICAL EVALUATION OF TABULAR Q-LEARNING VARIANTS

To isolate algorithmic mechanism from the confounds of neural function approximation, we evaluate four model-free value-based learners in the tabular setting, where the Q-function is an exact table and any performance difference is

TABLE XII
ATARI TASK CATEGORIES AS AXIS DIAGNOSTICS.

Task category	Diagnoses (axis)	Methods that excel	Representative finding
Reaction-time / dense reward	Function-approx stability (W8)	Double DQN, Dueling, PQN	Methods cluster within $\sim 2x$; overestimation control buys modest gains
Strategic planning	Credit assignment (W4)	QR-DQN, IQN, FQF	Distributional methods dominate Q*bert / Ms. Pac-Man
Sparse reward	Brittle exploration (W3)	NoisyNet, Bootstrapped DQN, DQfD, RND/Go-Explore	Intrinsic motivation necessary but not sufficient on Montezuma
Large observation	Function-approx stability (W8)	Dueling, distributional, PQN	Low inter-method variance; capability not the bottleneck

attributable to the update rule alone. The study is deliberately small and mechanism-focused; it is not a large-scale benchmark.

Protocol. We use three classic discrete environments — FrozenLake-v1 (4×4, slippery: stochastic transitions), Taxi-v3 (structured, near-deterministic), and CliffWalking (per-step cost with a cliff penalty) — and four learners: Q-learning, SARSA, Expected SARSA, and n -step Q-learning ($n = 3$). Each learner is trained over **100 random seeds** with an ε -greedy schedule decayed from 1.0 to 0.05; we then evaluate the greedy policy over 300 episodes and report the mean episodic return with a **95% bootstrap confidence interval** (10,000 resamples), following the interval-estimate practice advocated by [83]. The full configuration and a seeded, single-file reproduction script are provided as supplementary material.

Planning oracles, not competitors. We also report the optimal policy obtained by dynamic programming (Value and Policy Iteration). These methods require full access to the transition model P and reward R and so are *not* model-free competitors; they are reported only as a *planning upper bound* and are separated from the learners in the table accordingly.

What the results show. Three patterns, each illustrating an axis from §IV. (i) On *stochastic* FrozenLake, Q-learning and Expected SARSA (0.722, 0.720) nearly reach the 0.739 oracle, while n -step Q collapses to 0.497 with a wide CI: bootstrapping a multi-step return through stochastic transitions injects exactly the variance that the credit-assignment axis (§IV.D)(#sec-iv-d) trades against bias and the stability axis (§IV.H)(#sec-iv-h) warns of. (ii) On *near-deterministic* Taxi, all four learners converge to within 0.5% of the oracle with tight, overlapping CIs — when transitions are predictable the choice of update rule barely matters, a useful negative result. (iii) CliffWalking reproduces the canonical on-/off-policy contrast: *off-policy* Q-learning recovers the optimal cliff-edge path (−13.0, matching the oracle exactly, zero variance), whereas the expectation method prefers the *safe* route one row from the cliff (Expected SARSA −17.0); SARSA and n -step Q show large negative means and high variance because their sampled on-policy targets yield unstable greedy policies across seeds.

Scope. These tabular results are illustrative, not a leaderboard. They confirm that the axis distinctions of §IV manifest even without neural approximation, and that statistical rigor changes which differences are real: SARSA’s CliffWalking CI spans [−114, −51], so any single-seed comparison would be misleading. The same discipline — many seeds, interval

estimates rather than point scores — is what §V argues the historical Atari literature lacks.

VII. COMPARATIVE ANALYSIS OF DEEP Q-LEARNING REPOSITORIES

To assess how the methodological landscape of §IV is reflected in practice, we audited six widely-used open-source deep-RL repositories — Tianshou [12], XuanCe [13], CleanRL [14], DQN Zoo [15], Stable Baselines3 [16], and RLlib [17] — for Q-learning coverage, organized by the eight axes of §IV. The full coverage matrix (supplementary material) records each method against each repository; here we report the structure it reveals.

Coverage findings. Coverage is sharply skewed toward the early-era families. Overestimation methods (§IV.A)(#sec-iv-a), via Double and Dueling DQN) and distributional methods (§IV.D)(#sec-iv-d), via C51, QR-DQN, IQN, FQF) are the best-supported axes, an artifact of the 2016–2019 deep-RL boom during which most of these repositories were architected. Sample-efficiency (§IV.B)(#sec-iv-b) and exploration (§IV.C)(#sec-iv-c) are patchy — prioritized replay and HER appear in roughly half, NoisyNet in only one. The newer axes are largely absent: offline RL (§IV.E)(#sec-iv-e) is implemented natively by none of the six (practitioners defer to d3rlpy and similar), and multi-agent value decomposition (§IV.F)(#sec-iv-f) and distributed-scale methods (§IV.G)(#sec-iv-g) have no native VDN/QMIX/QPLEX or Ape-X/R2D2/Agent57 implementations — a structural gap, since these demand infrastructure beyond single-machine scope. Most pointed: nine methods surveyed in §IV — DRQN, CBDQ, DQfD, MeDQN, Bootstrapped DQN, UCB Q-Ensemble, EBQL, Posterior Sampling DQN, and PQN — are absent from *all six* repositories. This set overlaps heavily with the directions §IV flags as most promising, indicating that implementation availability is itself a bottleneck on adoption — the central motivation for the community-repository proposal in §VIII(#sec-viii).

Non-interchangeability. Coverage is necessary but not sufficient. Hundal et al. [90] show that different implementations of the *same* algorithm diverge materially in measured performance, because silent code-level choices — gradient clipping, observation and reward normalization, weight initialization — vary across repositories without appearing in the algorithm’s name. A “Double DQN” cell in one repository is therefore

TABLE XIII

TABULAR RESULTS: MEAN EPISODIC RETURN OVER 100 SEEDS WITH 95% BOOTSTRAP CIs. THE PLANNING ORACLE (FULL MODEL ACCESS) IS AN UPPER BOUND, NOT A MODEL-FREE COMPETITOR. [†]FROZENLAKE REPORTS SUCCESS RATE (MEAN REWARD, MAX 1.0).

Algorithm	FrozenLake-v1 [†]	Taxi-v3	CliffWalking
Q-learning	0.722 [0.701, 0.737]	7.93 [7.90, 7.96]	−13.0 [−13, −13]
SARSA	0.692 [0.669, 0.712]	7.90 [7.87, 7.93]	−79.9 [−114, −51]
Expected SARSA	0.720 [0.705, 0.732]	7.93 [7.90, 7.96]	−17.0 [−17, −17]
n -step Q ($n=3$)	0.497 [0.451, 0.541]	7.93 [7.89, 7.96]	−75.2 [−104, −46]
Planning oracle (VI/PI)	0.739	7.94	−13.0

TABLE XIV

PER-REPOSITORY STRENGTHS AND THE TRADE-OFFS THEY ACCEPT.

Repository	Strength	Trade-off
Tianshou	Broad coverage; best distributional/overestimation support	Reproducibility unverified across full suite
XuanCe	Modular YAML config; high code reuse	Defaults diverge from papers; steep curve
CleanRL	Single-file readability; easy to audit	Low modularity; compositional agents awkward
Stable Baselines3	Stable API; widely adopted	Few Q-learning variants beyond DQN
RLlib	Distributed scale out-of-the-box	Heavyweight; hard to debug per-algorithm
DQN Zoo	High reproduction fidelity to DeepMind	Scope limited to the DQN family

not directly comparable to the same cell in another. Cross-repository comparison consequently requires controlling for these implementation details rather than trusting nominal algorithm labels.

Named-algorithm vs. feature-equivalent support. Our audit marks a method supported only when a repository ships it as a *named* algorithm with exposed hyperparameters — the strictest reading. Feature-equivalent availability is higher: RLlib’s `DQNConfig`, for instance, exposes noisy, distributional, and prioritized-replay flags that compose into a Rainbow-like agent the matrix does not credit as a named-Rainbow implementation, an effect also present to a lesser degree in Tianshou and XuanCe.

Design trade-offs. The repositories occupy distinct points on a Pareto frontier rather than dominating one another. CleanRL [14] optimizes single-file readability, which limits modularity and makes compositional agents such as Rainbow awkward to express; DQN Zoo [15] optimizes reproduction fidelity at the cost of scope. At the other end, RLlib [17] optimizes distributed scale at the cost of debuggability and per-algorithm transparency, while Stable Baselines3 [16] optimizes API stability and broad adoption at the cost of Q-learning-variant breadth. Tianshou [12] and XuanCe [13] trade single-method depth for coverage breadth. Repository selection is thus itself an axis decision, and no single repository serves the Q-learning community comprehensively.

This audit is a dated snapshot, reflecting repository state as of 2026-05-01 (Tianshou 0.5.x, XuanCe 1.x, CleanRL master, DQN Zoo head, Stable Baselines3 2.x, RLlib 2.x); coverage evolves quickly and should be re-checked against

current releases.

VIII. CONCLUSION AND FUTURE WORK

A. Summary: This paper presents a problem-first survey of Q-learning and deep Q-learning, reorganizing three decades of methodological development around the eight foundational weaknesses of vanilla Q-learning introduced in §II.B: overestimation bias, sample inefficiency, brittle exploration, reward sparsity and credit assignment, distribution shift, multi-agent coordination, slow adaptation, and function-approximation instability.

For each weakness, §IV surveys the families of methods that respond:

- **§IV.A** — overestimation control via decoupling (Double Q, Double DQN) and ensembles (EBQL, REDQ).
- **§IV.B** — sample efficiency via prioritized sampling (PER), demonstration augmentation (DQd), memory-efficient consolidation (MeDQN), and goal relabeling (HER).
- **§IV.C** — directed exploration via noise injection (NoisyNet, Parameter Space Noise), ensemble disagreement (Bootstrapped DQN, UCB Q-Ensemble), belief modulation (CBDQ, PSDQN), and intrinsic motivation (RND, Go-Explore).
- **§IV.D** — credit assignment via multi-step returns and the distributional family (C51, QR-DQN, IQN, FQF).
- **§IV.E** — offline RL via policy constraint (BCQ, AWAC, BRAC), value penalty (CQL), expectile regression (IQL), and ensemble diversification (EDAC).
- **§IV.F** — cooperative value decomposition via additive (VDN), monotonic mixing (QMIX), duplex dueling (QPLEX), and constraint-free (QTRAN) factorizations.
- **§IV.G** — scaling and adaptation via distributed actor-learner architectures (Ape-X, R2D2), bandit-controlled exploration meta-policies (Agent57), and meta-learning on the Q-function.
- **§IV.H** — function-approximation stability via target networks, architectural decomposition (Dueling), normalization recipes (PQN), and consolidation losses (MeDQN, Munchausen DQN).

Sections V and VI present axis-stratified empirical evidence, showing that methods targeting weakness W_i excel on the task categories that diagnose W_i and remain near baseline elsewhere. Section VII analyzes algorithmic coverage across six open-source repositories, identifying nine methods absent from all surveyed codebases — a gap that motivates the principal forward-looking deliverable of this paper.

B. A community Q-learning repository: The repository coverage analysis of §VII surfaces a concrete implementation gap: nine methods covered in §IV — DRQN, CBDQ, DQfD, MeDQN, Bootstrapped DQN, UCB Q-Ensemble, EBQL, Posterior Sampling DQN, and PQN — are absent from *every* surveyed open-source repository. Across the new axes introduced in §IV.E, §IV.F, and §IV.G, the gap is broader still: most offline-RL, multi-agent value-decomposition, and distributed-Q methods have no implementation in any of the six repositories analyzed.

We propose the development of a **dedicated, community-maintained repository focused exclusively on Q-learning and its deep variants**. The repository would have four design priorities:

1. **Axis-aware coverage** — implementations organized by the eight weakness axes, prioritizing methods absent from existing repositories, with a coverage matrix maintained as the public roadmap.
2. **Unified training scripts** — a common interface across methods permitting like-for-like comparison under matched evaluation protocols.
3. **Standardized benchmarks** — Atari, classic control, D4RL, and SMAC included as canonical benchmark suites, with reference results for each implemented method.
4. **Empirical reproducibility audits** — each method release gated on reproduction of the original paper’s headline results, in the style of Hundal et al. [90].

The repository’s initial roadmap would prioritize the nine methods in §VII.C as the first implementation targets, followed by the offline-RL family (CQL, IQL, EDAC), the multi-agent value-decomposition family (QMIX, QPLEX, VDN), and the distributed-Q family (Ape-X, R2D2, Agent57). Cross-references between the repository’s documentation and this paper’s §IV sections would let practitioners navigate from “I face problem *X*” (this paper’s axis) to “available implementations addressing *X*” (the repository’s coverage matrix).

C. Open research directions, by axis: The “Open Questions” subsections (E) of each §IV axis-section collectively summarize the field’s frontier. We highlight four cross-axis directions of particular promise:

The compute-vs.-algorithmic tradeoff. The empirical pattern in §V suggests that distributed scaling (§IV.G) accounts for the largest single performance increase in deep RL since DQN, but PQN’s single-machine results suggest the algorithmic axes are far from saturated. A systematic study of which weaknesses are *compute-resolvable* versus *algorithmically-resolvable* — at fixed compute budgets — would clarify where research effort is best directed.

Cross-axis composition. Frontier agents (Agent57, NGU) explicitly compose mechanisms from multiple axes. A formal account of which axis combinations are synergistic, redundant, or antagonistic would convert the empirical pattern into a design principle. The Rainbow ablation [6] provides one such account on a smaller method set; a broader cross-axis composition study would extend it.

The offline-online interface. Methods bridging offline and online learning (AWAC, Cal-QL) address a regime increas-

ingly central to practical deployment but theoretically under-explored. The interface is itself a candidate weakness axis the field may recognize in coming years.

Q-learning under real-world distribution shift. §IV.E treats distribution shift as a *training* regime: the data was collected by one or more behavior policies and is then fixed. A deployment-side analog — the learned policy meets a state distribution different from the one it was trained on — is largely unaddressed. Sim-to-real transfer in robotic Q-learning, OOD robustness against perturbations, and online policy correction at deployment time all live in this regime. The newer benchmarks of §V.I (ProcGen for procedural variation, CARL for context generalization, RL Bench for sim-to-real) begin to provide evaluation infrastructure, but no unified Q-learning treatment of *deployment* distribution shift exists. This is the axis where the gap between Atari-evaluated methods and real-world deployment is most visible, and where Q-learning’s value-based perspective could productively engage with robotics, control, and operations-research literature that has developed parallel tools.

D. Limitations and Ethics: This is a survey rather than a deployed system, so the relevant considerations are scholarly and community-facing rather than direct model-harm concerns. We surface five limitations explicitly.

Benchmark monoculture. Tables II–III’s reliance on Atari and §VI’s tabular Gymnasium environments reflects the field’s historical convention. As documented in §V.H, Atari is a strong benchmark for a specific set of weaknesses (exploration, credit assignment, function-approximation stability) and a weak one for others (sample efficiency at small budgets, generalization across procedural variation, continuous control, real-world deployment). A survey that elevates Atari evidence to load-bearing status inherits these gaps. We have partly mitigated this through §V.I’s coverage of ProcGen, NetHack, BSuite, D4RL, and SMAC and through §IV.E and §IV.F’s adoption of D4RL and SMAC respectively, but the breadth of evaluation evidence remains uneven across the eight axes.

Compute inequality. Several of the methods covered (Agent57’s 78B training frames, R2D2’s 10B, large in-context Q-learning agents) require compute budgets accessible to only a handful of industrial labs. By including these methods alongside academically reproducible alternatives without flagging the gap, a survey risks implicitly endorsing a research culture where frontier work is structurally inaccessible to most readers. §IV.G’s discussion of the compute-versus-algorithmic trade-off and §V.H.5 on compute scale begin to address this, but the field-wide tension between *algorithmic discovery* and *compute-driven discovery* is underdeveloped here and warrants explicit treatment in future surveys.

Citation bias in narrative reviews. Any narrative survey that reduces a large field to “canonical methods” can inadvertently erase negative results, replication studies, and less-publicized alternatives — even when, as here, the source selection criteria are documented (§III.A). We provide the prior-art sweep and a full per-method method-type index as supplementary material — the audit trail for our differentiation claim and a bridge to the older organization — but we cannot eliminate the bias inherent in narrative selection. Readers

seeking a more systematic review-of-reviews should consult the Springer NCAA 2026 offline-RL distribution-shift survey and the broader-RL surveys of Ghasemi 2024/2025 and Murphy 2024/2025 as complementary navigations of overlapping terrain.

Stale tooling comparisons. The repository comparison of §VII is pinned to repository state as of early 2026. Implementation landscapes shift quickly; readers consulting this paper in 2027 or later should treat Table VII as a snapshot rather than current ground truth. The proposed Q-learning-specific repository (§VIII.B) is intended in part as a mechanism for maintained tooling comparison rather than as a one-shot artifact.

Frontier methods overstated relative to settled techniques. This paper devotes substantial space to in-context Q-learning (§IV.G.B.3), flow-matching policies (§IV.E.B.5), and recent theoretical advances (§IV.I.C) — all subject to ongoing community consensus formation. Some claims in those subsections may be revised as the field consolidates. We have flagged the relevant sections with explicit “active research area” language where applicable, but readers should treat any specific claim about post-2024 work as more uncertain than the well-established results in §IV.A–D and §IV.H.

A small note on broader ethical considerations: Q-learning’s expanding role in robotics, recommender systems, and language-model alignment carries downstream policy implications that this survey does not address. A reader interested in the responsible deployment of value-based RL methods should consult the broader RL-safety and RL-policy literature (Saunders et al. 2023, Anderljung et al. 2024) as complementary material.

E. Closing: Q-learning’s longevity as a foundational paradigm rests on its combination of conceptual simplicity and methodological extensibility. The thirty-five years since [Watkins 1989] have produced a tooling ecosystem rich enough to deserve organized treatment but specialized enough that no general-RL survey can capture its texture. This paper offers the first multi-axis problem-first synthesis of that ecosystem, paired with empirical and implementation analyses that support the structural claim.

The methodological landscape this paper traces is not closed. New axes will emerge — the offline-online interface and the meta-RL boundary are two candidates — and existing axes will continue to admit new mechanisms. The problem-first organization is intended to be extensible: new methods slot into existing axes, and new axes can be added as the field recognizes them. The framework’s value is precisely that it survives those additions.

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